

A Privacy-Preserving Distributed Training Architecture for Cyber Forensics Using Blockchain and Homomorphic Encryption

Abstract:

Cyber forensics plays a vital role in analyzing sensitive digital evidence where maintaining privacy, integrity, and secure collaboration is essential. Traditional centralized learning models pose significant risks, including data breaches, unauthorized access, and evidence tampering. To mitigate these challenges, this project introduces a Blockchain-Federated Learning Framework designed for secure, privacy-preserving training and aggregation of cyber forensic data. By allowing multiple agencies to collaboratively train models without sharing raw evidence, the framework enhances trust, transparency, and inter-agency cooperation. Existing works in Privacy-Preserving Federated Learning [1] and Blockchain-Based Federated Learning [2] have strengthened data security but fall short in addressing the specific needs of forensic data management and aggregation, underscoring the necessity for a domain-focused, transparent, and privacy-compliant solution.

The proposed framework integrates Federated Learning (FL), Blockchain, and Distributed Learning to achieve secure and decentralized collaboration across forensic nodes. Each participating agency trains its model locally and shares only encrypted parameters, while blockchain smart contracts verify node authenticity and immutably store model updates. Through homomorphic encryption and consensus-based validation, the system ensures confidentiality, prevents data leakage, and provides verifiable aggregation of updates. This decentralized approach strengthens data integrity, auditability, and reliability in cyber forensic intelligence sharing, creating a trustworthy ecosystem for secure collaboration among law enforcement and digital investigation agencies.

References: [1] Bin Gu., “Privacy-Preserving Asynchronous Vertical Federated Learning,” IEEE Trans. Neural Netw. Learn. Syst., vol. 33, no. 11, pp. 6103–6115, 2022.

[2] R. Yang et al., “Blockchain-Based Federated Learning with Enhanced Privacy,” IEEE Internet Things J., vol. 11, no. 12, pp. 21674–21688, 2024.

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